EE3001: Solid State Devices
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Assignment #6

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10}cm^{-3}$**;** k_BT**=0.026 eV (at room temperature);** $E_{G,Si}$ **= 1.12 eV and temperature independent; at 300 K,** N_c**=**$3.2 \times 10^{19}cm^{-3}$**,** N_v**=**$1.8 \times 10^{19}cm^{-3}$**,** $\mu_n = 1350cm^2/V.s$**,** $\mu_p = 450cm^2/V.s$**; Permittivity for vacuum:** ϵ_0**=**8.85×10^{-14}**F/cm, relative permittivity:** $\epsilon_{r,Si}$**=**11.8**,** ϵ_{r,SiO_2}**=**3.9**; pn diode dopants are completely ionized.**

1. (2 points) The conductance (g) versus forward biased current (I) plot for a pn diode with unity ideality factor yields the y-intercept and slope as 38.46×10^{-12} S and 38.46 V^{-1}, respectively.

   (a) Find out the bias voltage at which this diode yields 2 mA current.

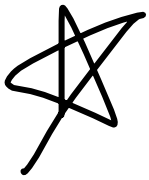
   (b) What is the diode resistance at this bias point?

2. (2 points) Given two identically and uniformly doped abrupt silicon p-n junction diodes, D1 and D2. Cross sectional area of D1 is twice of that of D2. The reverse saturation current for D1 is 10 nA.

   (a) Plot the forward current dependent conductance for both the diodes at room temperature (in the same figure) showing important projection points.

   (b) If both D1 and D2 were p^+-n diodes, plot their forward current dependent diffusion capacitance (in the same figure) if τ_p= 1 μs for both the diodes.

3. (5 points) Qualitatively draw the position (x) dependent electron and hole current densities ($J_n(x)$ and $J_p(x)$) for the following p-n diode structures where W_p, W and W_n represent the lengths of the quasi-neutral p-region, space-charge-region, and quasi-neutral n-region, respectively. Note that L_p and L_n denote the diffusion lengths for minority holes and electrons, respectively.

   (a) A p-n diode with $N_A > N_D$, $W_n \gg L_p$ and $W_p \gg L_n$.

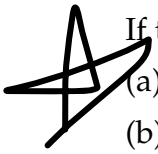
   (b) A p-n diode with $N_A > N_D$, $W_n \ll L_p$ and $W_p \ll L_n$.

   (c) A p-n diode with $N_A > N_D$, $W_n \gg L_p$ and $W_p \ll L_n$.

   (d) A p^+-n diode with a $W_n \gg L_p$.

   (e) A p^+-n diode with a $W_n \ll L_p$.

4. (5 points) For a forward biased p^+-n diode, the hole current density ($J_p(x)$) at $x = x_n$ is found to be 0.1 mA/cm^2. Given that τ_p = 1 μs.



If the length of the quasi-neutral n-region is 300 μm, answer the following:

(a) What is the total diode current density at the given bias?

(b) What is the electron current density, $J_n(x)$ at $x = x_n + 2$ μm?

(c) What is the total stored excess hole charge in the quasi-neutral n-region?

If the length of the quasi-neutral n-region is 2 μm, answer the following:

(d) What is the total stored excess hole charge in the quasi-neutral n-region?

(e) What is the electron current density, $J_n(x)$ at $x = x_n + 1$ μm?